The Subtractive Prime-Gate Problem

DEFINITIONS

Throughout, n is an integer with $n \geq 2$. Let $P(n)$ denote the largest prime factor of n. Define a map D on $\{2, 3, 4, \ldots\}$ by

$$D(n) = \begin{cases} 3n + 1 & \text{if } n \text{ is prime,} \\ n - P(n) & \text{if } n \text{ is composite.} \end{cases}$$

(For composite n the value $n - P(n)$ is again at least 2, so D maps $\{2, 3, 4, \ldots\}$ into itself.) Write D^k for the k-fold iterate of D, with $D^0(n) = n$.

THE CYCLE C

D has the following cycle, which we call C (note $D(14) = 7$, closing the loop):

$$7 \to 22 \to 11 \to 34 \to 17 \to 52 \to 39 \to$$
$$26 \to 13 \to 40 \to 35 \to 28 \to 21 \to 14 \to 7$$

PROBLEM

Prove or disprove: for every integer $n \geq 2$, the orbit of n eventually enters C; that is, there exists $k \geq 0$ such that $D^k(n) \in C$.

Equivalently: C is the only cycle of D, and no orbit of D is unbounded.